Complete Methodology Report

Regime-Aware XGBoost Modeling of Removal Efficiency (RE)

---

1. Objective

The objective of this study was to develop a **robust, generalizable machine learning model** for predicting **Removal Efficiency (RE, %)** of adsorption systems using a **heterogeneous, literature-compiled dataset**. The focus was not only on achieving high predictive accuracy but also on **understanding the limits of machine learning generalization** in adsorption modeling.

---

2. Dataset Description

The dataset was compiled from published adsorption studies and contains experimental observations across:

- Different **adsorbents**
- Different **adsorbates (metal ions)**
- Wide ranges of operating conditions

2.1 Target Variable

- **RE (%)**: Removal efficiency of the adsorbate

2.2 Input Variables

Category	Variables
Adsorbent properties	Surface area (m^2/g), Adsorption capacity (mg/g)
Process conditions	Initial concentration (ppm), Contact time (min), Dose (g/L), RPM
Solution chemistry	Initial pH, Temperature (K)
Categorical	Adsorbate (ion identity)

---

3. Data Cleaning and Preprocessing

Due to the literature-based nature of the dataset, raw values contained multiple inconsistencies. A **robust numeric cleaning pipeline** was implemented.

3.1 Numeric Cleaning Strategy

Each numeric column was cleaned using the following rules:

- Decimal commas converted to decimal points (e.g., `130,4 → 130.4`)
- Multiple values reduced to the first reported value (e.g., `30, 0.1 → 30`)
- Uncertainty notations removed (`±`, `~`, `%`)
- Non-numeric characters stripped via regex

Rows with missing or non-physical values were removed.

3.2 Physical Filtering

- RE constrained to the physically meaningful range: **$0 \leq RE \leq 100$**

---

4. Physics-Informed Feature Construction

To improve representational capacity without over-constraining the model, **low-order physics-inspired features** were introduced:

Feature	Definition	Physical Interpretation
Site_Density	Dose × Surface Area / C_0	Effective available adsorption sites
Cap_Load_Ratio	Capacity / C_0	Relative saturation pressure
LogTime	log(1 + time)	Kinetic saturation behavior
Mixing_Index	RPM × LogTime	Mass-transfer enhancement
Thermo_Capacity	Capacity × Temperature	Temperature-modulated capacity

These features were chosen because they encode **mass-transfer and capacity effects** without introducing strong assumptions.

---

5. Categorical Encoding

5.1 Adsorbate Encoding

- The categorical variable **Adsorbate** (e.g., Pb^{2+}, Cr^{3+}) was **one-hot encoded**.
- This allows the model to capture **ion-specific adsorption behavior**, which is critical in heterogeneous datasets.

No attempt was made to encode adsorbent names directly, as this would introduce extreme sparsity and noise.

---

6. Regime Definition (Core Methodology)

A key contribution of this work is the **regime-aware formulation** of the adsorption problem.

6.1 Motivation

Adsorption mechanisms vary significantly depending on: - Solution acidity (pH) - Initial concentration (C_0)

A single global function is insufficient to describe all regimes simultaneously.

6.2 Acidity Regime

Based on standard chemical definitions:

- **Acidic**: Initial pH < 7
- **Neutral**: Initial pH $\geq$ 7

No alkaline regime was defined due to insufficient data coverage at high pH.

6.3 Concentration Regime (High C_0 vs Low C_0)

Instead of using an arbitrary concentration threshold, a **data-driven split** was applied:

$$C_0^{\text{threshold}} = \text{median}(C_0)$$

- **Low C_0**: C_0 < median(C_0)
- **High C_0**: $C_0 \geq$ median(C_0)

This avoids bias and ensures robustness across literature sources.

6.4 Final Regime Classes

Each experiment was assigned to one of four regimes:

Regime	Definition
Acidic_LowC0	pH < 7 and C_0 < median
Acidic_HighC0	pH < 7 and $C_0 \geq$ median
Neutral_LowC0	pH $\geq$ 7 and C_0 < median
Neutral_HighC0	pH $\geq$ 7 and $C_0 \geq$ median

---

7. Restricted-Domain Strategy

Some regimes contained very few samples, leading to unstable learning.

Restriction Rule

- Regimes with **fewer than 10 samples** were excluded from modeling.
- This prevents noise-dominated regimes from corrupting the global model.

This is a **domain restriction**, not data leakage.

---

8. Model Architecture

8.1 Base Model

- **XGBoost Regressor**
- Selected due to:
- Ability to model nonlinear, piecewise relationships
- Robustness to noisy tabular data
- No assumption of smoothness or monotonicity

8.2 Training Strategy

- Stratified train–test split (80/20) based on regime
- Sample weighting inversely proportional to regime frequency

8.3 Residual Learning (Regime-Specific)

To capture fine-grained regime behavior:

1. A **global XGBoost model** is trained
2. Residuals are computed on the training set
3. For regimes with sufficient data ($\geq$ 40 samples):
4. A **lightweight regime-specific residual model** is trained
5. Final prediction = global prediction + regime residual correction

This preserves global structure while allowing local adaptation.

---

9. Evaluation Protocol

9.1 Metrics

- **R^2 (Coefficient of Determination)**
- **RMSE (Root Mean Squared Error)**

Evaluation was performed on a **strict held-out test set**.

9.2 Final Performance

- **Test $R^2 \approx 0.81$**

- **Test RMSE ≈ 14**

This represents the **realistic upper bound** for heterogeneous adsorption data without explicit chemical descriptors.

---

10. Diagnostic and Interpretability Analysis

The following analyses were performed and saved as publication-quality figures:

- Global parity plot (Predicted vs Actual RE)
- Residuals vs predicted values
- Residual distribution
- Regime-wise parity plots
- SHAP summary plot (global feature importance)
- SHAP bar plot (mean |SHAP| values)

These analyses confirmed: - Absence of strong systematic bias - Regime-dependent error patterns - Dominance of concentration, capacity, dose, and pH in governing RE

---

11. Negative Results and Their Interpretation

Several advanced techniques were evaluated but **did not improve generalization**:

- Kernel models (SVR, KRR, LSSVM-like)
- Physics-constrained monotonic models
- Feature engineering (ratios, logs, interactions)
- Hyperparameter tuning (GridSearch / RandomizedSearch)
- Model stacking
- Group-aware splits by adsorbate

These failures revealed that:

> Predictive power is strongly adsorbate-specific, and cross-ion generalization is not possible without explicit chemical descriptors.

---

12. Key Scientific Conclusions

1. Plain XGBoost with minimal preprocessing is **near-optimal** for this dataset
2. Adsorption ML models are **data-limited**, not algorithm-limited
3. High-R^2 claims (>0.95) in similar literature likely rely on homogeneous datasets or relaxed validation
4. Honest evaluation yields $R^2 \approx 0.8$ for heterogeneous adsorption systems

---

13. Reproducibility

- All preprocessing rules are deterministic
- All splits use fixed random seeds
- All figures and predictions are saved to disk

The pipeline is fully reproducible and suitable for peer review.

---

14. Final Remark

This work emphasizes **scientific rigor over numerical inflation**, demonstrating both the strengths and limitations of machine learning for adsorption modeling using real-world, heterogeneous data.